

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Weekly Progress Report — Week 4

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1 Introduction

Week 4 marks a significant milestone for the Fabrix project. Having established the foundational dataset loading and image preprocessing pipelines in Week 3, our efforts this week shifted toward completing the full implementation of the “Fixed Pipeline” (Phase 1). This involved extracting multi-scale texture features (LBP, GLCM, Gabor), generating deviation maps, and rigorously evaluating the system’s performance on the AITEX dataset to empirically understand the limitations of classical fixed-parameter approaches.

2 Fixed Pipeline Implementation

We successfully completed the implementation of the Fixed Pipeline, which utilizes pre-determined, non-adaptive spatial scales for texture analysis. During this phase, we processed the entire AITEX dataset consisting of 245 high-resolution (4096×256) images, extracting thousands of non-overlapping patches for localized analysis.

2.1 Texture Descriptor Extraction

Building upon the preliminary code from last week, we finalized the integration of the classical feature extraction modules:

- **LBP (Local Binary Patterns):** Finalized the extraction of local micro-texture histograms, computing the spatial patterns that define the baseline texture profile.
- **GLCM Features:** Completed the extraction of six key Haralick features (contrast, homogeneity, energy, ASM, correlation, and dissimilarity) calculated over local windowed regions at multiple orientations to accurately capture woven thread patterns.
- **Gabor Filter Banks:** Implemented multi-scale, multi-orientation frequency and spatial texture analysis to detect variations in texture periodicity across the fabric patches.
- **Deviation Mapping & Feature Fusion:** Developed the mathematical framework to calculate Z-scores for every extracted feature against a normal-texture reference (fitted purely on defect-free training images). We then implemented a weighted feature fusion module to combine these per-descriptor maps into a single unified anomaly probability map.

2.2 Intelligent Defect-Region Validation

A major challenge encountered during initial testing was balancing the system’s sensitivity to small defects against its tendency to generate false alarms. Initially, a hard morphological cutoff (minimum component area of 500 pixels) artificially reduced false alarms but severely damaged the system’s ability to detect small, yet significant, true defects.

To resolve this, we engineered an **Intelligent Defect-Region Validation** module within our connected component localization pipeline (`src/localization/connected_components.py`). We lowered the minimum area filter to 25 pixels and introduced a dynamic shape-and-confidence filter. This enhancement dynamically penalizes regions based on area and shape, successfully rejecting structural shadows and normal weave variations while preserving small, highly-anomalous defects.

3 Evaluation and Insights

We conducted rigorous empirical testing of the Fixed Pipeline on our validation splits to understand the mathematical boundaries of the classical fixed-scale approach.

3.1 Performance Observations

By tuning our thresholding parameters to a hypersensitive state ($local_k = 1.2$), we forced the Fixed Pipeline to detect defects across all 7 fabric types (ending the “blindness” problem on difficult textures). At this setting, the system successfully caught every defective image in the validation set.

However, this came at the cost of generating a massive amount of false alarms. While true defects were successfully recovered across challenging fabrics, background noise was repeatedly flagged as anomalous. By lowering the threshold, we successfully localized anomalous pixels on fabrics that previously had exactly 0 detections (e.g., Fabrics 00, 03, 04, 05, 06), but the background noise exploded.

3.2 The Sensitivity Tradeoff

These results perfectly validate our core hypothesis and the classical sensitivity tradeoff inherent to fixed-scale texture descriptors. Traditional inspection methods (like fixed Gabor Filters) are optimized for a specific weave size and are mathematically incapable of adapting to diverse textures. When fixed filters tuned for Fabrics 01 and 02 are applied to other fabrics, they fail. Adjusting thresholds to artificially force detections only results in unacceptable levels of false positives.

4 Work Completed This Week

- Integrated LBP, GLCM, and Gabor filter banks into the fully functional Fixed Pipeline.
- Implemented intelligent morphological validation to dynamically filter components by shape and confidence rather than strict size cutoffs.
- Executed comprehensive thresholding evaluations of the Fixed Pipeline on the AITEX dataset.
- Analyzed output metrics to empirically prove the limitations of fixed texture parameters across diverse fabrics.

5 Plan for Next Week

Having proven the limitations of the Fixed Pipeline, next week will focus entirely on developing our proposed solution: the **Adaptive Pipeline (Phase 2)**.

- Implement 2D Autocorrelation to mathematically estimate the base texture scale of different fabrics directly from the images.
- Dynamically derive Gabor and LBP parameters based on the estimated fabric scales.
- Run the Adaptive Pipeline and begin the comparative analysis against the Fixed Pipeline.

End of Week 4 Report